FeelMySparks 748i

Project status, experimental FBW model, and next steps

Prepared 15 October 2026 · X-Plane 12 · Development summary

At a glance: The project is an evolving 747-8I-inspired aircraft built on a Sparky-derived 747-400 foundation. The inherited aircraft systems remain the baseline. The current FBW work is an experimental, partial augmentation in Lua—not a complete or certified 747-8 flight-control system.

Aircraft and project model

The project aims to develop a Boeing 747-8 Intercontinental-inspired aircraft for X-Plane 12. It currently builds on a Sparky-derived 747-400 aircraft and its existing XTLua systems, hydraulic controls, and autopilot. The aircraft's 747-8I exterior, engines, wings, cockpit, and flight-model changes are a longer-term development goal; their completion should not be inferred from the FBW work.

The FBW module is intentionally integrated as a separate Lua module inside the existing XTLua plugin. It adjusts selected control ratios at the existing hydraulic mixer stage and leaves the inherited hydraulic and autopilot callbacks in place. This limits integration risk and allows the master FBW DataRef to disable the augmentation.

Work completed

- **FMC presentation:** Custom alert-color work added magenta/cyan overlays while retaining amber alerts. A display refresh issue was addressed by preparing full text and writing changed strings only. The display backing transparency was later restored.
- **Weight and Balance plugin:** The standalone custom plugin was removed from the aircraft package and stale setup documentation was cleaned up. This does not imply that a new custom W&B tool replaced it.
- **FBW integration:** A standalone-in-the-plugin Lua module was added to the existing hydraulic mixer. An XTLua loader behavior that discards a loaded script's return value was accounted for by publishing the module API into the shared namespace.
- **Autopilot compatibility:** Roll-rate damping is suppressed while the aircraft's autopilot servos are engaged, so the damping law will not

deliberately oppose AP roll commands.

- **Documentation and copies:** FBW documentation has been updated in the aircraft and project READMEs. The module was synced across the development, Intercontinental, backup, and installed X-Plane aircraft copies.

Current experimental FBW model

FBW law inputs are sampled at **20 Hz** (every 0.05 seconds). The existing hydraulic mixer still applies the results during its normal update. The current prototype includes:

Law / feature	Current behavior and limits
Low-speed roll response	Speed-faded outboard-aileron differential boost; default gain 0.05, clamped to 0–0.15, fading to zero by 220 KIAS.
Roll-rate damping	Capped roll-rate opposition near neutral aileron demand, up to ±6 mixer units. Default gain 0.25. Active only airborne and with AP servos disengaged.
Flap-linked droop	Outboard-aileron common-mode droop follows flap ratio, up to 2.5 mixer units.
Maneuver-load alleviation	Prototype normal-G schedule begins above 1.3 G and approaches full demand near 2.1 G. Output is slew-limited at the 20 Hz cadence; it adjusts outboard ailerons and selected spoiler ratios.
Stick shaker cue	Uses X-Plane's native force-feedback stick-shaker DataRef as a stall-warning cue where supported. It is not simulated airframe buffet.

High-speed outer-aileron lockout	Existing hydraulic-model behavior is retained; it is separate from the added FBW laws.

These are tunable simulation approximations, not validated Boeing laws. The module does **not** command elevator or rudder, protect against a hand-flown stall, inhibit rotation below FMC Vr, model OAMS, or measure actual gust loads. Pitch and yaw remain with the inherited aircraft controls. The module's name and update frequency do not make the aircraft fly-by-wire in the full-aircraft sense.

Next work

1. **Evaluate the 20 Hz change in flight.** Compare the feel against the prior cadence, paying attention to turn response, turbulence, small manual inputs, and whether faster law sampling makes a noticeable difference.
2. **Retune from observed behavior.** If it still feels slow, record speed, weight, flap setting, AP state, and the specific maneuver. Adjust the responsible gain or hydraulic response only after isolating the cause; update cadence alone does not guarantee more aircraft response.
3. **Check FBW interactions.** Exercise the AP damping bypass, MLA, flap droop, and the inherited high-speed lockout at safe altitude and across representative configurations. Confirm the FBW master-off path restores baseline behavior.
4. **Improve testability.** Capture relevant DataRefs and simulator logs during flight tests, and add repeatable checks for the control-law math and copy parity where practical.
5. **Develop the 747-8I aircraft itself.** Plan and implement the exterior wing, engine nacelle and pylon work; review cockpit modernization and flight-model changes separately from the FBW prototype.
6. **Resolve distribution prerequisites.** Confirm permissions, credits, and license compatibility for inherited and third-party assets before packaging or redistribution.
7. **Release preparation.** Update installation and limitations documentation, test in the target X-Plane version, and distinguish verified behavior from experimental or planned features.

Useful FBW controls

- `laminar/B747/fbw/enabled` — FBW master; set to 0 to disable the augmentation.
- `laminar/B747/fbw/roll_response_gain` — low-speed roll-response gain.
- `laminar/B747/fbw/roll_rate_damping_gain` — manual-flight roll-rate damping gain.
- `laminar/B747/fbw/mla_enabled`, `droop_enabled`, and `buffet_enabled` — feature enables.
- `laminar/B747/fbw/mla_demand` and `aileron_droop_ratio` — output/diagnostic values.

Testing note: The user has been flight-testing the aircraft and reported that the handling is working well but feels slow; the law update rate was therefore increased from 5 Hz to 20 Hz. The faster cadence should be judged in the next simulator test. Do not treat this experimental aircraft as suitable for real-world aviation.

This status report summarizes development work and known limitations. It is not an engineering substantiation, certification document, or representation of Boeing documentation.